PATIENT COMMUNICATION AGENT – TECHNICAL PACKAGE
Includes:
• UML Class Diagrams (text)
• JSON Schemas
• Data Flow Diagrams (DFD)
• MongoDB NoSQL Model
• Snowflake/BigQuery Warehouse Model
• HIPAA Architecture Diagram
• OpenAPI API Specification
• Sequence Diagrams
• Real Demo Data

UML CLASS DIAGRAM (Mermaid)
classDiagram
class Patient {
+UUID patient_id
+string first_name
+string last_name
+date dob
+string gender
+string phone
+string email
}
Patient --> MessageHistory

JSON API SCHEMA (Messages)
{
"message_id": "uuid",
"patient_id": "uuid",
"text": "I need help",
"timestamp": "2024-01-01T10:00:00Z"
}

DATA FLOW DIAGRAM (Level 0)
Patient → Agent → NLU Engine → LLM → DB → Patient

MONGODB MODEL (Messages)
{
"_id": "uuid",
"patient_id": "uuid",
"text": "I need to reschedule",
"intent": "appointment_reschedule"
}

SNOWFLAKE MODEL (fact_messages)
message_id STRING
patient_id STRING
intent_id STRING

HIPAA ARCHITECTURE
Public Zone → API Gateway → App Layer → PHI Zone DB → Audit Logs → IAM/WAF/SIEM

OPENAPI SPEC (Messages)

paths:
/messages:
post:
summary: Send a message
requestBody:
required: true
responses:
200:
description: Message stored

SEQUENCE DIAGRAM
Patient → Chatbot: Sends message
Chatbot → NLU Engine: Classify intent
NLU → LLM: Generate safe response
LLM → Chatbot: Return response
Chatbot → Patient: Deliver reply

REAL DEMO DATA
Patients:
1) John Doe (ID: P001)
2) Sarah Lee (ID: P002)
Messages:
- P001: "I need a refill"
- P002: "When are my test results ready?"